Sr. No#	Technique	Type of Data			Paper/ref	Code	Application areas	Desc.	Association/Buzz words	Year	Type of model	Category	Notes
		Most	Mid	Least									
1	Layer-wise relevance propagation (LRP)	outdoor classification,	tabular, spatial-temporal	indoor object recognition, Synthetic-aperture radar (SAR), IR	https://journals.pl	https://git.tu-berli	DNN (object recognition + object seg), RNN (LSTM)	uses the network weights and the neural activations created by the forwa	global explanations, backward perturbations,	2015	Model Agnostic	Method (XAI)	https://iphome.hhi.de/samek/pdf/MonXAI19.pdf https://www.mdpi.com/2076-3417/12/1/136 https://www.sciencedirect.com/science/article/pii/S0925231219316613 https://www.mdpi.com/1424-8220/21/13/4536 https://www.sciencedirect.com/science/article/pii/S0378778822004881 https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12534/1253416/Quantized-deep-learning-for-low-cost-IR-sensors-with-layer/10.1117/12.2663748.short
2	Counterfactual method (counterfactual exp) (LIME, CEP etc)				https://arxiv.org/p	https://paperswithcode.com/task/counterfactual-explanation				2017			
	Counterfactual visual explanations (more appropriate match)	object recognition,	xray segmentation,	faces,	https://arxiv.org/p	https://github.con	GANs, DNN	what region in the image made the model predict class c instead of class c 0 ?	where and why,	2022	Model Agnostic	Technique	https://link.springer.com/article/10.1007/s10994-023-06333-w https://www.ijcai.org/proceedings/2020/0742.pdf
3	Generalized additive model (GAM)		tabular		https://pdodds.w	https://paperswitl	DNN	a generalized linear model in which the linear response variable depends	Statistical model, fit data relatio	1986	Model Agnostic	Method (XAI)	https://montrealethics.ai/game-changer-or-not-an-evaluation-of-interpretable-machine-learning-models/ https://arxiv.org/pdf/2006.06466v2.pdf
4	Accumulated Local Effects (ALE)		tabular		https://academic.	https://docs.seldo	CNN	describe how features influence the prediction of a machine learning mo	Statistical model, fit data relatio	2016	Model Agnostic	Visualization	
5	Class Activation Mapping (CAM)	outdoor classification, object segmentation	xray segmentation,	remote sensing, IR	https://openacce	https://github.con	DNN,	lets us see which regions in the image were relevant to this class.	class relevance, weakly-superv	2016	Model Agnostic	Method (XAI), AM	https://www.mdpi.com/1424-8220/20/17/4750 https://ieeexplore.ieee.org/abstract/document/10288391
6	Gradient-weighted Class Activation Mapping (Grad-CAM)	object classification	spatial-temporal	x-ray,	https://openacce	https://github.con	DNN, RL,	to understand which regions of an input image are important for the netw	class relevance	2017	Model Agnostic	Method (XAI), AM	https://link.springer.com/chapter/10.1007/978-3-031-48415-5_1
7	Peak Response Maps (PRM)	outdoor classification, object segmentation	x-ray	remote sensing	https://arxiv.org/p	https://github.con	DNN	identify individual instances and utilizes peak response cues together to	instance masks	2018	Model Agnostic	AM	
8	CLass-Enhanced Attentive Response (CLEAR)	object classification	tabular		https://arxiv.org/p	N/A	DNN	mitigates some of the shortcomings of heatmap-based methods associa	attention mechanism	2017	Model Agnostic	Method (XAI)	
9	Deep Resolve	x-ray	synthetic		https://icmlviz.git	https://github.con	DNN	an advanced reconstruction technology	visualizes intermediate layer feature maps	2017	Model Agnostic	Method (XAI)	
10	SCOUTER	medical imaging	object classification		https://arxiv.org/a	https://github.con	DNN	transparent classification with slot-attention	attention mechanism	2020	Model Agnostic	Method (XAI)	
11	Visual Feedback	object classification			https://arxiv.org/p	https://webserver	DNN	average visualizations of reduced set of features	class relevance	2019	Model Specific	Technique	
12	SHAP	object recognitiopn, Textual	tabular	remote sensing, Synthetic-aperture radar (SAR)	https://arxiv.org/a	https://github.con	ML, DNN, RNN, Transf	aims to explain any model by considering each feature (or predictor) in th	scoring feature importance	2017	Model Agnostic	Method (XAI)	
13	LIME	text, object classification	tabular	time-series,	https://www.kdd.c	https://github.con	DNN, RNN	regional justifications for predictions provided by sophisticated machine l	why, class relevance	2016	Model Agnostic	Method (XAI)	https://proceedings.mlr.press/v139/garreau21a.html
14	GNNExplainer	graph datasets			https://arxiv.org/a	https://github.con	GNN	identifies a compact subgraph structure and a small subset of node featu GNN's prediction	node relevance	2019	Model Agnostic	Method (XAI)	
15	GraphLIME	graph datasets			https://arxiv.org/a	https://github.con	GNN	non-linear explanation method for GNN	node relevance	2020	Model Agnostic	Method (XAI)	
16	Excitation Backpropagation (EB)	object classification			https://arxiv.org/a	https://github.co	DNN, RNN	gradient on the original image can then be loosely interpreted as the prol	class relevance	2018	Model Agnostic	Method (XAI)	https://www.researchgate.net/publication/329748634_Excitation_Backprop_for_RNNs
17	Attention Mechanism (esp for exp)	object classification, object recognition, textual			https://proceedin	https://machinele	DNN, RNN, Transform	assigns importance scores to each input feature, providing an explanatic	scoring feature importance	2017	Model Specific	Technique	
18	GNN-LRP	graph datasets			https://ieeexplore	https://git.tu-berli	GNN	a collection of walks into the input graph that are relevant for the predicti	node relevance	2021	Model Agnostic	Method (XAI)	
19	Saliency	object detection, object recognition, outdoor classification	x-ray	remote sensing	https://arxiv.org/a	https://github.con	DNN	reflects the degree of importance of a pixel to the human visual system	scoring feature importance	2013	Model Agnostic	AM	
20	Occlusion Sensitivity	object detection, object recognition, outdoor classification	x-ray	remote sensing	https://arxiv.org/p	https://github.con	DNN	which parts of an image are most important for a deep network's classific	network sensitivity	2013	Model Agnostic	Technique	(check date)
21	Concept Activation Vectors	xray, object classification		3D-generated data,	https://arxiv.org/p	https://github.con	DNN	a way for machines to represent images (what we understand) using nur	global explanations, neural con	2018	Model Specific	Technique	https://openaccess.thecvf.com/content/CVPR2022W/DLGC/html/Druc_Concept_Activation_Vectors_for_Generating_User-Defined_3D_Shapes_CVPRW_2022_paper.html
22	Partial Dependence Plots (PDP)		tabular		https://projecteuc	https://scikit-learr	ML, DNN	figures dependence between the target response and a set of input featu	class relevance	2001	Model Agnostic	Visualization	https://link.springer.com/article/10.1007/s10994-023-06327-8
23	Individual Condition Expectations plots (ICE)		tabular		https://arxiv.org/p	https://scikit-learr	ML, DNN	dependence between the target function and an input feature of interest	feature importance	2014	Model Agnostic	Visualization	
24	Leave One Column Out (LOCO)		tabular	text	https://arxiv.org/p	https://github.con	ML, DNN		feature importance	2016	Model Agnostic	Technique	https://www.frontiersin.org/articles/10.3389/fnbot.2021.784514/full
25	Accumulated Local Effects (ALE)		tabular		https://arxiv.org/p	https://github.con	ML, DNN	actual conditional marginal distributions instead of considering each mar	conditional distributions	2019	Model Agnostic	Visualization	
26	Anchors	object recognition	tabular		https://homes.cs.	https://github.con	ML, DNN	explains individual predictions of any black box classification model by fir	feature importance	2018	Model Agnostic	Method (XAI)	
27	Deep Learning Important Features (DeepLIFT)	object detection, object recognition, outdoor classification		sequential	https://arxiv.org/a	https://github.con	DNN	compares the activation of each neuron to its 'reference activation' and a	class relevance	2017	Model Agnostic	Method (XAI)	
28	Contrastive Explanations Method (CEM)	object detection, object recognition,	tabular, x-ray		https://arxiv.org/p	https://github.com/SeldonIO/alibi/blob/m		find what should be minimally and sufficiently present (viz. important obje to justify its classification and analogously what should be minimally and	present and absent	2018	Model Agnostic	Method (XAI)	
29	ProfWeight				https://arxiv.org/a	https://github.com/Trusted-AI/AIX360/blob/master/examples/profwt/Prof-Weight.ipynb				2018	Model Agnostic		For information tarnsfer (not xAI)
30	Permutation Feature Importance (model reliance)		tabular		https://arxiv.org/a	https://scikit-learr	ML, DNN	measures the increase in the prediction error of the model after we perm	feature importance	2018	Model Agnostic	Method (XAI)	https://link.springer.com/article/10.1007/s10994-023-06385-y
31	AI Explainability 360 (AIX360)				https://www.jmlr.c	https://github.com/Trusted-AI/AIX360				2020	Model Agnostic	Tool/Framework	
32	Skater				N/A	https://pypi.org/project/skater/					Model Agnostic	Tool/Framework	just a package/ documentation not available anymore
33	Explain Like I'm Five (ELI5)				https://arxiv.org/a	https://explainlikeimfive.io/				2019	doesnot explain /	Tool/Framework	
34	InterpretML				https://arxiv.org/a	https://interpret.ml/				2019	Model Agnostic	Tool/Framework	
35	Activation Atlases	object detection, object recognition			https://blog.resea	https://paperswitl	DNN	using feature inversion to visualize millions of activations from an image (	feature inversion	2019	Model Agnostic	Method (XAI)	
36	Alibi Explain				https://www.jmlr.c	https://github.com/SeldonIO/alibi				2021	Model Agnostic	Tool/Framework	
37	What-if Tool (WIT)				https://arxiv.org/a	https://pair-code.github.io/what-if-tool/				2019	Model Agnostic	Tool/Framework	
38	azureml-interpret				N/A	https://pypi.org/project/azureml-interpret/					Model Agnostic	Tool/Framework	just a pacakge
39	Rulex Explainable AI				https://www.scier	https://www.rulex.ai/native-explainable-ai/		https://github.com/ElsevierSoftwareX/SOFTX-D-22-00098		2022	Model Agnostic	Tool/Framework	
40	DALEX - Model Agnostic Language for Exploration and Explanation				https://www.jmlr.c	https://dalex.drwhy.ai/python/				2021	Model Agnostic	Tool/Framework	
41	Integrated Gradients	object detection, object recognition,	x-ray	Point Cloud,	https://arxiv.org/a	https://github.con	DNN, RNN	allows a class-wise attribution of the predictions with respect to the input	class relevance, attribution	2017	Model Specific	Method (XAI)	https://www.mdpi.com/1999-4893/16/7/316
42	XGNN	graph datasets			https://arxiv.org/a	https://github.con	GNN	high-level insights and generic understanding of how GNNs work	node relevance	2020	Model Agnostic	Method (XAI)	
43	ShapFlow	graph datasets	tabular		https://proceedin	https://github.con	DNN, ML	extends the flat, set-based, view prevalent in game theory based explana	casual links	2021	Model Agnostic	Method (XAI)	
44	Asymmetric Shapley Values (ASV)	tabular			https://arxiv.org/pdf/1910.06358.pd		ML	relax the symmetry axiom, in order to place more importance on causal a	casual explanations	2021	Model Agnostic	Method (XAI)	Theory* https://thesis.eur.nl/pub/67853/BSc_thesis_Erasmus_School_of_Economics_Sophie_Gehrels_Final_Version.pdf
45	Deep Taylor Decomposition (DTD)	object classification	x-ray	remote sensing	https://pdf.scienc	https://github.con	DNN	exploits the property that the function learned by a deep network is decoi	recursive	2017	Model Agnostic	Method (XAI)	https://arxiv.org/pdf/2211.08425.pdf
46	Prediction Difference Analysis (PDA)	object classification	x-ray		https://arxiv.org/a	https://github.con	DNN	method highlights areas in a given input image that provide evidence for	class relevance	2017	Model Agnostic	Visualization	
47	EXplainable Neural-Symbolic Learning (X-NeSyL)	data + Knowledge Graph			https://arxiv.org/a	https://paperswitl	DNN	designed to learn both symbolic and deep representations, together with the level of alignment of machine and human expert explanations	Neurosymbolic	2021	Model Specific	Method (XAI)	primarily for object detection tasks, uses SHAP GED and SHAP Backprop
48	Meaningful Perturbations	object classification	x-ray		https://openacce	https://github.con	DNN	general framework for learning different kindsn of explanations for any bl	Iterated gradients	2017	Model Agnostic	Method (XAI), AM	
49	Protodash	object classification, object segmentation			https://www.sem	https://github.con	ML, DNN	method for selecting prototypical examples that capture the underlying d	What-if	2017	Model Agnostic	Method (XAI)	requires a separate subset for explanations
50	Global Interpretation via Recursive Partitioning (GIRP)	object classification, textual			https://arxiv.org/p	https://github.con	DNN	creates binary tree based on class probabilities	inherent explanations	2018	Model Agnostic	Method (XAI)	weak representation
51	Diverse Counterfactual Explanations (DICE)		tabular		https://dl.acm.org	https://github.con	ML	supports generating a set of counterfactual explanations and has tunable	What-if	2020	Model Agnostic	Method(XAI)	
52	InputGradient	object classification	tabular	textual	https://arxiv.org/a	https://github.con	DNN, ML	studies the partial derivatives of the model with respect to the input	gradient mapping	2016	Model Specific	AM	https://arxiv.org/abs/1312.6034
53	SmoothGrad	object detection, object recognition			https://arxiv.org/a	https://github.con	DNN	method that can help visually sharpen gradient-based sensitivity maps, a	gradient mapping	2017	Model Agnostic	AM	
54	NoiseGrad	object detection, object recognition			https://arxiv.org/a	https://github.con	DNN	improves attribution-based explanations by introducing stochasticity to th	gradient mapping	2021	Model Agnostic	AM	
55	FusionGrad	object detection, object recognition			https://arxiv.org/a	https://github.con	DNN	improves attribution-based explanations by introducing stochasticity to th	gradient mapping	2021	Model Agnostic	AM	same paper as NoiseGrad
56	CXPlain	object detection, object recognition	textual		https://arxiv.org/a	https://github.con	ML, DNN, RNN	learns to estimate to what degree certain inputs cause outputs in anothe	attribution, casual explanations	2019	Model Agnostic	Method (XAI)	
57	DeepDream	object detection, object recognition			https://blog.resea	https://www.tens	DNN	find and enhance patterns in images via algorithmic pareidolia, thus crea	What explanations	2015	Model Agnostic	Method (XAI)	

58	Network Dissection	object detection, object recognition			https://arxiv.org/p	https://github.con	DNN	works by measuring the alignment between unit response and a set of co	latent representations	2017	Model Agnostic	Method (XAI)		
59	ANOVA		tabular		https://dl.acm.org	https://scikit-learn	ML (post-DNN testing)	measure for how much variance each feature is responsible in the model	feature importance	2004	Model Agnostic	Technique	Statistical function	
60	AtMan	video data, textual			https://arxiv.org/a	https://github.con	Transformers	modality-agnostic perturbation method that manipulates the attention me	attention mechanism	2023	Model Agnostic	Method (XAI)		
61	Spectral relevance analysis (SpRAy)	object classification, object detection			https://www.natu	https://www.natu	DNN	way of characterizing and validating the behavior of nonlinear learning m	heatmap analysis	2019	Model Agnostic	AM	all code sources are linked in this paper	
62	InputIBA	object classification, object detection	textual		https://arxiv.org/a	https://github.con	DNN	a method to identify features with predictive information in the input doma	attribution, latent representation	2021	Model Agnostic	Method (XAI)		
63	Information bottleneck attribution (IBA)	object classification, object detection			https://openrevie	https://github.con	DNN	adding noise to intermediate feature maps to restrict the flow of informati	attribution	2020	Model Agnostic	Method (XAI)	(published in 2020, was out in 2019) 2020	
64	SimplEx	object classification, object detection			https://arxiv.org/p	https://github.con	DNN	a user-centred method that provides example-based explanations with re	reconstruction, latent represent	2021	Model Agnostic	Method (XAI)		
65	OmniXAI				https://www.sema	https://github.com/salesforce/OmniXAI				2022	Model Agnostic	Tool/Framework		
66	ExplainX.ai				N/A	https://github.com/explainX/explainx				2020	Model Agnostic	Tool/Framework	just a pacakge	
67	XRAI	object classification	x-ray		https://openacces	https://github.con	DNN	first over-segments the image, then iteratively tests the importance of ea	attribution, class relevance	2019	Model Agnostic	Method (XAI), AM		
68	Xplique				https://arxiv.org/p	https://github.com/deel-ai/xplique				2022	Model Agnostic	Tool/Framework		
69	CARLA				https://arxiv.org/p	https://github.com/carla-recourse/CARLA				2021	Model Agnostic	Tool/Framework	just a pacakge	
70	Gradient Activation Maps (GAM)	object classification, object detection, segmentation	x-ray		https://arxiv.org/a	https://github.con	DNN	uses localized gradient and activation information from multiple network l	gradient mapping	2021	Model Agnostic	AM		
71	OpenXAI				https://arxiv.org/a	https://open-xai.github.io/				2022	Model Agnostic	Tool/Framework		
72	Concept-based XAI (CXAI)				https://arxiv.org/p	https://github.com/dmitrykazhdan/concept-based-xai				2021	Model Agnostic	Tool/Framework		
73	NeuroCartography	object classification, object detection			https://arxiv.org/p	https://github.con	DNN	an interactive system that scalably summarizes and visualizes concepts	neural concepts	2021	Model Agnostic	Tool/Framework, AM		
74	TorchEsegeta				https://arxiv.org/p	https://github.com/soumickmj/TorchEsegeta				2022	Model Agnostic	Tool/Framework		
75	Quantus				https://jmlr.org/pa	https://github.com/understandable-machine-intelligence-lab/quantus/				2023	Model Agnostic	Tool/Framework		
76	Interpretable Convolutional Neural Networks	object classification, object detection			https://arxiv.org/p	https://github.con	DNN	used to clarify knowledge representations in high conv-layers of CNNs	feature importance	2018	Model Agnostic	Technique		
77	Grad Cam++	object classification	spatial-temporal	x-ray,	https://arxiv.org/p	https://github.con	DNN	better object localization as well as explaining occurrences of multiple ob	class relevance	2018	Model Agnostic	Method (XAI)		
78	Interpretable Basis Decomposition	object classification, object detection			https://openacces	https://github.con	DNN	provides an explanation for a prediction by decomposing the decision in	feature inversion	2018	Model Agnostic	Technique		
79	Full-Grad	object classification, object detection			https://arxiv.org/p	https://github.con	DNN	obtained by aggregating the full-gradient components, which decompose	gradient mapping	2019	Model Agnostic	Method (XAI)		
80	This Looks Like That (ProtoPNet)	object classification, object detection			https://proceedin	https://github.con	DNN	the network dissects the image by finding prototypical parts, and combin	attention mechanism	2019	Model Agnostic	Method (XAI)		
81	sufficient input subsets	object classification, object detection	textual		https://proceedin	https://github.con	DNN, RNN	identify minimal subsets of features whose observed values alone suffice	feature importance	2018	Model Agnostic	Technique		
82	Explanatory Interactive Machine Learning (XIL) (CAIPI)	object classification, object detection	textual		https://arxiv.org/p	https://codeocea	ML, DNN	learner explains its query to the user, and the queries of any active classi	active learning, LIME	2020	Model Agnostic	Tool/Framework, Technique		
83	Contextual Semantic Interpretability (CSIB)	object classification, object detection	remote sensing		https://arxiv.org/p	https://github.con	DNN	connects the semantic bottleneck with the final output by using non-linea	semantic attribution	2020	Model Specific	Method (XAI)		
84	Conceptual Explanation	object classification, object detection	textual, time series	remote sensing	https://ieeexplore	N/A	DNN, RNN	makes use of the fact that many relevant concepts for time series can inc	expert knowledge		Model Agnostic	Technique		
85	Interpnet	object classification			https://arxiv.org/a	https://paperswit	DNN	relies on the assumption that the network's computation and reasoning is	natural language based explana	2017	Model Agnostic	Method (XAI)		
86	INNvestigate Neural networks				https://arxiv.org/a	https://github.com/albermax/innvestigate/				2018	Model Agnostic	Tool/Framework		
87	explAIner				https://arxiv.org/a	https://github.com/dbvis-ukon/explainer				2019	Model Agnostic	Tool/Framework		
88	Exmo	object recognition	textual		https://arxiv.org/p	N/A	DNN	consists of a list of IF...THEN... statements with a decision rule in the con	feature inversion	2022	Model Agnostic	Method (XAI)		
89	MeLime	object classification, object detection	textual		https://arxiv.org/p	https://github.con	ML, DNN	generalizes the LIME method, allowing more flexible perturbation sampli	why, class relevance	2020	Model Agnostic	Method (XAI)		
90	M4				https://proceedin	https://github.con	DNN, Transformers			2023	Model Agnostic	Tool/Framework		
91	eXplego				https://nr.brage.u	explego.nr.no				2023	Model Agnostic	Tool/Framework		
92	CHILLI	tabular, textual	object recognition		https://wrap.warv	N/A	ML, DNN, RNN	method for incorporating data context into a post-hoc XAI framework whe	feature importance	2023	Model Agnostic	Technique		
93	CRAFT	object classification, object detection			https://openacces	https://github.con	DNN	uses a recursive formulation to identify concepts at the correct level of gr	what and where	2023	Model Agnostic	Method (XAI)		
94	VERIX	object classification, object detection			https://proceedin	https://github.con	DNN	builds such explanations and counterfactuals iteratively using constraint-	what and why	2023	Model Agnostic	Method (XAI)		
95	Explora	graph data			https://dl.acm.org	https://github.con	RL	synthesizes network-oriented explanations based on an attributed graph	node relevance	2023	Model Specific	Method (XAI)		
96	R-cut	object classification, object detection			https://arxiv.org/p	N/A	Transformers	module performs fine-grained feature decomposition, taking into account	fetaure relevance	2023	Model Specific	Method (XAI), AM		
97	SyntaxShap	textual			https://arxiv.org/a	N/A	LM	extends Shapley values to account for parsing-based syntactic depender	semantic attribution	2024	Model Agnostic	Method (XAI)		
98	TempMe	graph data			https://proceedin	https://github.con	GNN	extracts the most interaction-related motifs while minimizing the amount	node relevance	2023	Model Agnostic	Method (XAI)		
99	D4explainer	graph data			https://proceedin	https://github.con	GNN	approach that provides in-distribution GNN explanations for both counter	node relevance	2023	Model Agnostic	Method (XAI)		
100	Concept Relevance Propagation (CRP)	object classification, object detection			https://arxiv.org/p	https://github.con	DNN	ndicates not only which input features are relevant, but also communicat	.where and why	2024	Model Agnostic	Method (XAI)		
101	Tame	object classification			https://arxiv.org/p	https://github.con	DNN	method for generating explanation maps with a multi-branch hierarchical	attention mechanism	2023	Model Agnostic	Method (XAI), AM		
102	AttnLRP	video data, textual			https://arxiv.org/p	N/A	Transformers	faithfully and holistically attribute not only input but also latent representa	LRP, attention mechanism	2024	Model Specific	Method (XAI)		
103	AtttnRoll	video data, textual			https://aclantholo	https://github.con	Transformers	post hoc method when we use attention weights as the relative relevance	attention mechanism	2020	Model Specific	Method (XAI)		
104	VISCUIT	object classification			https://openacces	https://poloclub.g	DNN	visually summarizes the subgroups on which the classifier underperform	bias explanations, neural conce	2022	Model Agnostic	Tool/Framework, AM		10
105	ScrutinAI	object classification			https://diglib.eg.org/xmlui/bitstream		DNN	includes interactive visualizations of the model's inputs and outputs, inter	neural concepts	2022	Model Agnostic	Tool/Framework, AM		
106	TSXplain	time series			https://arxiv.org/p	N/A	DNN	uses the extracted statistical features to describe the decision of a NN, m	natural language based explana	2019	Model Agnostic	Method (XAI)		
107	RISE (Randomized Input Sampling for Explanation)	object classificati	time series, x-ray		http://bmvc2018.	https://github.con	DNN	estimates importance empirically by probing the model with randomly ma	saliency based	2018	Model Agnostic	Method (XAI)		